

Gaourav Patil

Backend Engineer | Python | DevOps | Kubernetes | CI CD

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Professional Summary

RHCSA-Red Hat Certified System Administrator with hands-on experience in **DevOps, CI/CD pipeline automation, containerization (Docker, Kubernetes), and AWS Cloud** infrastructure, combined with strong Linux administration and scripting (Python, Bash). Experienced building scalable microservices, REST APIs, and applying secure coding and DevSecOps-aligned practices across the delivery lifecycle. Passionate about integrating AI-assisted automation into CI/CD workflows and IT operations to improve reliability and engineering efficiency. Comfortable in Agile teams, collaborating with developers, architects, and QA to drive delivery and platform stability.

Skills

CI/CD & Automation	Jenkins, GitLab CI (exposure), Git, GitHub, CI/CD Pipeline Design, GitOps
Containers & Orchestration	Docker, Kubernetes, Microservices Architecture
Cloud & Infrastructure	AWS, Google Cloud Platform (GCP), Terraform, Cloud Networking, Linux Administration (RHCSA)
Scripting	Python, Bash, Go, Java, C++
DevSecOps	Secure Coding, JWT Auth, Secrets Management (exposure)
APIs & Monitoring	REST API Design, Structured Logging, Postman, Prometheus, Grafana, Observability
Databases	PostgreSQL, MongoDB, Supabase
Frameworks	Express, React, Flask, FastAPI
AI Tools	Claude AI, Cursor, ChatGPT, Vercel v0
Methodologies	Agile, Cross-functional Collaboration, Code Review
Languages	English, Hindi, Marathi

Projects

Nexus – Cross Model Persistent Memory GitHub
Individual

- Developed an **LLM orchestration platform** providing Memory as a Service (MaaS) that enables multiple AI models to share persistent contextual memory across sessions.
- Engineered a cross-model memory layer allowing models such as GroqAI, MistralAI, ChatGPT, and Gemini to exchange summarized context, reducing redundant token consumption.
- Achieved a **45% reduction in token usage** through intelligent context summarization, memory retrieval, and optimized prompt management, forming the foundation of a scalable SaaS architecture.
- Built and deployed the backend using Go, Supabase, and PostgreSQL, provisioning cloud infrastructure with **AWS, Docker, and Terraform**, and integrating multiple LLM providers with a modular orchestration layer for efficient AI workflow management.

Technologies: Python, React, Gemini, Groq AI, Mistral AI, Claude, AWS, Docker, Terraform, Supabase

Guardrail – Self-Healing CI/CD Pipeline for Kubernetes GitHub
Individual

- Built a **self-healing CI/CD pipeline** for **Kubernetes** delivering zero-downtime deployments with automated health checks and rollback, removing manual intervention on failed releases.
- Designed the pipeline using **GitHub Actions** with a self-hosted runner, building multi-stage **Docker** images (~12.1MB final size) and pushing SHA-tagged images to GitHub Container Registry (GHCR).
- Implemented **Kubernetes rolling updates** with readiness/liveness probes and an independent Bash health-check gate; automated rollback via `kubectl rollout undo` achieved **0 seconds downtime** on failed deploys.
- Instrumented the app with **Prometheus** metrics and **Grafana** dashboards for real-time **observability** into request rate and latency during deployments.

Technologies: Go, Docker, Kubernetes, GitHub Actions, Prometheus, Grafana, GHCR, Bash, CI/CD

AI DevOps Orchestrator GitHub
Individual

- Developed an AI-enabled **DevOps orchestration platform** for **Kubernetes** resource management through REST APIs and a centralized execution engine, supporting **CI/CD** and GitOps-style workflows.
- Implemented **JWT authentication**, dependency injection, structured logging, and standardized API responses using secure coding principles.
- Built a scalable **microservices architecture** supporting AI-assisted deployments and cloud-native automation.

Technologies: Node.js, Express.js, React.js, Python, PostgreSQL, REST APIs, GroqAI, Kubernetes

Education

MIT World Peace University, Pune Jul 2025 – Jul 2027
Master's of Computer Application (MCA), CGPA: 7.29

- Developed and deployed Nexus – Cross Model Persistent Memory as a practical engineering project.

Rajarshi Shahu Mahavidyalaya, Latur Jul 2022 – Jul 2025
Bachelor's of Computer Application (BCA), CGPA: 6.67

Certifications: RHCSA – Red Hat Certified System Administrator; Google Associate Cloud Engineer (In Progress)